Offline POMDP Algorithms

Break

So far, we have only dealt with n-step value functions, if we want to deal with infinite-horizon discounted POMDPs, how can we construct an infinite-horizon α vector to start with?

Infinite-Horizon POMDP Lower Bound Improvement

$\Gamma \leftarrow$ blind lower bound

loop

$\quad \Gamma \leftarrow \Gamma \cup \mathrm{backup}(\Gamma)$

$\quad \Gamma \leftarrow \mathrm{prune}(\Gamma)$

Complexity $O\left(|\Gamma|\,|A|\,|O|\,|S|^2 + |A|\,|S|\,|\Gamma|^{|O|}\right)$

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$\mathrm{backup}(\Gamma)$

$$\alpha^{a,o}[s] = \sum_{s'} Z(o \mid a, s')\, T(s' \mid s, a)\, \alpha[s'] \quad \forall\, a, o, s, \alpha \in \Gamma$$

$$\Gamma^{a,o} = \left\{ \frac{1}{|O|}\, r_a + \alpha^{a,o} \;:\; \alpha \in \Gamma \right\} \quad \forall\, a, o$$

$$\Gamma^{a} = \bigoplus_{o \in O} \Gamma^{a,o} \quad \forall\, a$$

$$\Gamma' = \bigcup_{a \in A} \Gamma^{a}$$

Complexity $O\big(|\Gamma|\,|A|\,|O|\,|S|^2 + |A|\,|S|\,|\Gamma|^{|O|}\big)$

Point-Based Value Iteration (PBVI)

Maintain belief set B, repeatedly call point_backup for each $b \in B$

point_backup(Γ, b)

for $a \in A$

for $o \in O$

$b' \leftarrow \tau(b, a, o)$

$\alpha_{a,o} \leftarrow \underset{\alpha \in \Gamma}{\operatorname{argmax}}\ \alpha^\top b'$

for $s \in S$

$\alpha_a[s] = R(s, a) + \gamma \sum_{s',o} T(s' \mid s, a)\, Z(o \mid a, s')\, \alpha_{a,o}[s']$

return $\underset{\alpha_a}{\operatorname{argmax}}\ \alpha_a^\top b$

$$O\big(|\Gamma|\,|A|\,|O|\,|S|^2 + |A|\,|S|\,|\Gamma|\,|B|\big)$$

Original PBVI

$B \leftarrow b_0$

loop

  for $b \in B$

    $\Gamma \leftarrow \Gamma \cup \{\text{point_backup}(\Gamma, b)\}$

  $B' \leftarrow \emptyset$

  for $b \in B$

    $\tilde{B} \leftarrow \{\tau(b, a, o) : a \in A, o \in O\}$

    $B' \leftarrow B' \cup \left\{ \underset{b' \in \tilde{B}}{\operatorname{argmax}} \, \|B, b'\| \right\}$

  $B \leftarrow B \cup B'$

Heuristic Search Value Iteration (HSVI)

while $\overline{V}(b_0) - \underline{V}(b_0) > \epsilon$

  explore$(b_0, 0)$

explore(b, t)

  if $\overline{V}(b) - \underline{V}(b) > \epsilon\gamma^t$

    $a^* = \underset{a}{\operatorname{argmax}}\ \overline{Q}(b, a)$

    $o^* = \underset{o}{\operatorname{argmax}}\ P(o \mid b, a)\left(\overline{V}(\tau(b, a^*, o)) - \underline{V}(\tau(b, a^*, o)) - \epsilon\gamma^t\right)$

    explore$(\tau(b, a^*, o^*), t + 1)$

    $\underline{\Gamma} \leftarrow \underline{\Gamma} \cup \text{point_backup}(\underline{\Gamma}, b)$

    $\overline{V}(b) = B_b\left[\overline{V}(b)\right]$

Sawtooth Upper Bounds

SARSOP

Successive Approximation of Reachable Space under Optimal Policies

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Policy Graphs

Monte Carlo Value Iteration (MCVI)

MC-Backup (G, b, N)

$R_a = 0 \quad V_{a,o,v} = 0$

for $a \in A$

for i in $1:N$

$s_i \leftarrow \text{sample}(b)$

$s'_i, o_i, r_i \leftarrow G(s_i, a)$

$R_a + r_i$

for $v \in G$

$V_{a,o_i,v} = V_{a,o_i,v} + \text{Simulate}(G, v, s'_i, L)$

for o in O

$V_{a,o} = \max_{v \in G} V_{a,o,v}$

$v_{a,o} = \operatorname{argmax}_{v \in G} V_{a,o,v}$

$V_a = R_a + \gamma \sum_o V_{ao}/N$

$V^* = \max_a V_a$

$a^* = \operatorname{argmax} V_a$

add new node to G labeled with a^*